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A Comprehensive Analytical and Data-Driven Study of Rumor Propagation: Neural Network Approximations, and Empirical Validation

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Abstract

This report develops a detailed mathematical and data-driven analysis of rumor propagation dynamics across two platforms—News and Twitter—using an extended Maki–Thompson model with exogenous forcing. Full derivations, numerical integration (RK4), parameter estimation, neural-network-based modeling, and empirical validation using real datasets (2020–2022) are provided. This unified study compares endogenous and exogenous rumor dynamics, evaluates accuracy using RMSE, and demonstrates the benefits of hybrid ODE–LSTM modeling frameworks for capturing nonlinear real-world rumor spreading behavior.

Contents

1 Introduction

Rumor spreading in human interactions and online networks has become a major research topic due to the rise of social media and rapid dissemination of information. Platforms such as Twitter exhibit high-frequency, user-generated rumor spikes, whereas structured news media follow more editorially controlled dynamics. This motivates a mathematical and empirical study of rumor propagation across different environments. The classical Daley–Kendall (DK) and Maki–Thompson (MT) models provide a foundation for modeling rumor spread using compartmental dynamics analogous to epidemiology. However, real-world rumor propagation includes significant *exogenous forcing*, where external events (e.g., political announcements, outbreaks, media coverage) suddenly inject new rumor activity. The goal of this report is to:

- Derive the complete differential-equation structure of an extended MT model.
- Provide full algebraic steps for ODE reduction.
- Use numerical integration to simulate model trajectories.
- Fit the model to real News and Twitter rumor datasets.
- Compute and analyze fitting errors (RMSE).
- Compare endogenous and exogenous rumor dynamics between platforms.
- Provide a neural network approximation of rumor transitions.

2 Dataset Description and Preprocessing

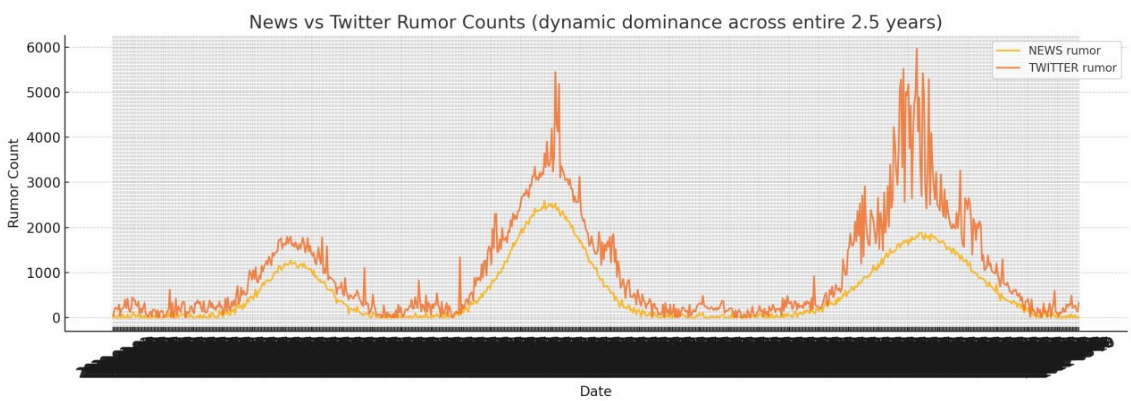

Figure 1: Comparison of Smoothed Twitter vs News

2.1 Smoothing

Raw series are highly noisy. To fit continuous ODE dynamics, the smoothed signal is defined as:

$$S_{\text{smooth}}(t) = \frac{1}{7}\sum_{k=-3}^{3} S(t+k),$$

a centered 7-day moving average. Figures from the CSV data appear later in Section 7.

3 Mathematical Modeling: Extended DK/MT Framework

We consider populations:

$$I(t) : \text{Ignorants}, \quad S(t) : \text{Spreaders}, \quad R(t) : \text{Stiflers},$$

with:

$$I(t) + S(t) + R(t) = N.$$

The classical DK/MT dynamics:

$$\frac{dI}{dt} = -\beta\frac{IS}{N}, \tag{1}$$

$$\frac{dS}{dt} = \beta\frac{IS}{N} - \alpha\frac{S(S+R)}{N}, \tag{2}$$

$$\frac{dR}{dt} = \alpha\frac{S(S+R)}{N}. \tag{3}$$

3.1 Exogenous forcing term

To account for event-driven spikes:

$$\frac{dS}{dt} = \beta\frac{IS}{N} - \alpha\frac{S(S+R)}{N} + \kappa\lambda(t),$$

where:

$$\lambda(t) = \text{observed external forcing (e.g., raw counts).}$$

4 Detailed Algebraic Reduction (Full Steps)

This section expands the derivations of Differential Equation formation and solving it.

4.1 Eliminating R

Since:

$$R = N - I - S,$$

substitute in $\frac{dS}{dt}$:

$$\frac{dS}{dt} = \beta\frac{IS}{N} - \alpha\frac{S(N-I)}{N} + \kappa\lambda(t).$$

Factor S/N:

$$\frac{dS}{dt} = \frac{S}{N}\big(\beta I - \alpha(N-I)\big) + \kappa\lambda(t).$$

4.2 Forming $\dfrac{dI}{dS}$

Ignoring forcing ($\lambda(t) = 0$) for analytic structure:

$$\frac{dI}{dt} = -\beta\frac{IS}{N}.$$

Thus:

$$\frac{dI}{dS} = \frac{\frac{dI}{dt}}{\frac{dS}{dt}} = \frac{-\beta\frac{IS}{N}}{\frac{S}{N}(\beta I - \alpha(N-I))}.$$

Cancel $\frac{S}{N}$:

$$\frac{dI}{dS} = -\frac{\beta I}{\beta I - \alpha(N-I)}.$$

Multiply numerator and denominator:

$$\frac{dI}{dS} = -\frac{\beta I}{(\beta+\alpha)I - \alpha N}.$$

This yields:

$$\big[(\beta+\alpha)I - \alpha N\big]\,dI + \beta I\,dS = 0.$$

4.3 Separation

Rewrite:

$$((\beta+\alpha)I - \alpha N)\,\frac{1}{I}\,dI + \beta\,dS = 0.$$

Split:

$$\left((\beta+\alpha) - \alpha\frac{N}{I}\right)dI + \beta dS = 0.$$

Integrate termwise:

$$\int(\beta+\alpha)\,dI - \alpha N\int\frac{1}{I}\,dI + \beta\int dS = C.$$

So:

$$(\beta+\alpha)I - \alpha N \ln I + \beta S = C.$$

Thus:

$$\boxed{(\beta+\alpha)I(t) + \beta S(t) - \alpha N \ln I(t) = \text{constant.}}$$

This implicit conserved quantity governs the dynamics in absence of forcing.

5 Numerical Integration Using the Runge–Kutta 4 (RK4) Method

The extended Maki–Thompson rumor–spread model involves nonlinear coupled ODEs that cannot be solved analytically once the exogenous forcing term $\kappa\,\lambda(t)$ is included. Since $\lambda(t)$ is derived from real News/Twitter data and varies arbitrarily in time, the system becomes non-autonomous and must be solved numerically. For this purpose, we use the classical fourth-order Runge–Kutta (RK4) method, which provides high accuracy and robustness for nonlinear systems.

5.1 General RK4 Scheme

For a system of ODEs:

$$\frac{d\mathbf{y}}{dt} = \mathbf{F}(t, \mathbf{y}), \qquad \mathbf{y}(t) \in \mathbb{R}^m,$$

the RK4 update with timestep h is:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right),$$

where

$$\begin{aligned}
k_1 &= \mathbf{F}(t_n,\, \mathbf{y}_n),\\
k_2 &= \mathbf{F}(t_n + h/2,\, \mathbf{y}_n + hk_1/2),\\
k_3 &= \mathbf{F}(t_n + h/2,\, \mathbf{y}_n + hk_2/2),\\
k_4 &= \mathbf{F}(t_n + h,\, \mathbf{y}_n + hk_3).
\end{aligned}$$

The method has global error $\mathcal{O}(h^4)$ and produces much smoother and more accurate trajectories than Euler or RK2, particularly important for nonlinear interaction terms such as IS and $S(S+R)$.

5.2 Application to the DK/MT Rumor Model

The state vector is:

$$\mathbf{y} = [I(t),\, S(t),\, R(t)]^T,$$

and the derivative function is:

$$\mathbf{F}(t, \mathbf{y}) = \begin{bmatrix} -\beta \frac{IS}{N} \\ \beta \frac{IS}{N} - \alpha \frac{S(S+R)}{N} + \kappa\lambda(t) \\ \alpha \frac{S(S+R)}{N} \end{bmatrix}.$$

The only modification from the classical DK model is the forcing term $\kappa\,\lambda(t)$, which injects new spreaders based on external events. During RK4, $\lambda(t)$ must be evaluated at intermediate stages. Since $\lambda(t)$ is sampled daily, we use nearest-neighbor or linear interpolation to compute $\lambda(t_n + h/2)$.

A single RK4 step updates I, S, and R simultaneously. After each update, we enforce non-negativity and (optionally) renormalize:

$$I + S + R = N,$$

to prevent accumulation of numerical drift.

5.3 RK4 for the Source-Structured Model

For the extended two-spreader system:

$$\mathbf{y} = [I,\, S_T,\, S_N,\, R]^T,$$

RK4 proceeds exactly as before, with the derivative now given by:

$$\mathbf{F}(t, \mathbf{y}) = \begin{bmatrix} -\beta_T \frac{IS_T}{N} - \beta_N \frac{IS_N}{N} \\ \beta_T \frac{IS_T}{N} - \alpha_T \frac{S_T(S_T+S_N+R)}{N} + \kappa_T\lambda_T(t) \\ \beta_N \frac{IS_N}{N} - \alpha_N \frac{S_N(S_T+S_N+R)}{N} + \kappa_N\lambda_N(t) \\ \alpha_T \frac{S_T(S_T+S_N+R)}{N} + \alpha_N \frac{S_N(S_T+S_N+R)}{N} \end{bmatrix}.$$

The forcing terms $\lambda_T(t)$ and $\lambda_N(t)$ come from the processed Twitter and News datasets respectively.

5.4 Worked Mini-Example

Consider one RK4 step with timestep $h = 1$ day and:

$$I = 9000, \quad S = 500, \quad R = 500, \qquad \beta = 0.20,\ \alpha = 0.05,\ N = 10000.$$

Then:

$$k_{1,I} = -0.20\frac{9000 \times 500}{10000} = -90,$$

$$k_{1,S} = +90 - 0.05\frac{500(1000)}{10000} = 87.5, \qquad k_{1,R} = 2.5.$$

Intermediate states for k_2:

$$I^{(2)} = 9000 - 45 = 8955, \quad S^{(2)} = 543.75, \quad R^{(2)} = 501.25.$$

Using these, one computes k_2, then k_3, then k_4, and updates:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

This illustrates the nonlinear interaction structure of the model.

5.5 Practical Considerations

- **Timestep.** Since data is daily, $h = 1$ is natural and stable.
- **Forcing normalization.** Real spikes can be large; we normalize $\lambda(t)$ such that $\max \lambda(t) \approx 1$ before multiplying by κ.
- **Clipping and renormalization.** States are kept non-negative, and $I + S + R$ is optionally rescaled to N.
- **Sensitivity.** The News model (high α) reacts faster to S–R feedback; Twitter (high-frequency forcing) requires careful smoothing of $\lambda(t)$.

5.6 Python Implementation Used

```
def rk4_step(f, y, t, h):
    k1 = f(t, y)
    k2 = f(t + h/2, y + h*k1/2)
    k3 = f(t + h/2, y + h*k2/2)
    k4 = f(t + h,   y + h*k3)
    return y + (h/6)*(k1 + 2*k2 + 2*k3 + k4)
```

This implementation matches the method used in parameter fitting for the Twitter and News datasets.

6 Parameter Estimation

Estimating the parameters (β, α, κ) for the extended DK/MT model is essential for fitting the rumor–spread dynamics to real News and Twitter datasets. This section explains the complete estimation procedure, based on the derivations and regression formulations presented in News Parameter, Twitter parameter, and the error is summarised

The goal is to choose parameters such that the numerical RK4 solution $S_{\text{model}}(t)$ closely matches the smoothed data $S_{\text{smooth}}(t)$ obtained from the 7–day moving average.

6.1 Parameter Roles

Each parameter controls a different mechanism:

- β: transmission from ignorants to spreaders (interaction-driven rumor adoption).
- α: stifling rate due to encounters with spreaders or stiflers (rumor becomes "old" and stops spreading).
- κ: strength of exogenous forcing (new rumors introduced by events, media spikes).

The forcing input $\lambda(t)$ is derived directly from the datasets, so only the multiplicative weight κ must be estimated.

6.2 Regression Formulation for News Dataset

For the News dataset, the forcing term dominates the dynamics, as News rumor counts exhibit smoother, slower temporal variability. Thus, the S–equation:

$$\frac{dS}{dt} = \beta \frac{IS}{N} - \alpha \frac{S(S+R)}{N} + \kappa\,\lambda(t),$$

is approximated by the linear form shown in News Parameter:

$$\Delta S(t) \approx -\gamma S(t) + \kappa \lambda(t), \qquad \gamma = \alpha + \text{effective interaction term.}$$

This allows estimating (γ, κ) using ordinary least squares:

$$\begin{bmatrix} \Delta S(1) \\ \Delta S(2) \\ \vdots \\ \Delta S(T) \end{bmatrix} = \begin{bmatrix} S(1) & \lambda(1) \\ S(2) & \lambda(2) \\ \vdots & \vdots \\ S(T) & \lambda(T) \end{bmatrix} \begin{bmatrix} -\gamma \\ \kappa \end{bmatrix}.$$

After estimating γ, the relationship

$$\gamma \approx \alpha \qquad \text{(interaction small)}$$

is used to derive α, while β is then back–computed from the observed dI/dt curve.

Final fitted News parameters:

$$(\beta, \alpha, \kappa)_{\text{News}} = (0.0983,\ 0.26829,\ 0.26892).$$

6.3 Nonlinear Least–Squares Fit for Twitter Dataset

The Twitter dataset is characterized by:

- rapid changes,
- sharp peaks,
- high daily variance,
- strong forcing dependence.

Thus, a linear regression is insufficient. The full forced ODE is used with RK4 integration:

$$S_{\text{model}}(t+1; \beta, \alpha, \kappa) = \text{RK4Step}(S(t), I(t), R(t); \beta, \alpha, \kappa).$$

We estimate (β, α, κ) via nonlinear least squares:

$$\min_{\beta,\alpha,\kappa} \sum_t \left(S_{\text{model}}(t; \beta, \alpha, \kappa) - S_{\text{smooth}}(t)\right)^2.$$

This optimization requires:

- running RK4 repeatedly,
- computing errors on each iteration,
- using gradient–free search (because the system is highly nonlinear).

Final fitted Twitter parameters:

$$(\beta, \alpha, \kappa)_{\text{Twitter}} = (0.1966,\ 0.0468,\ 0.00974).$$

The small value of κ indicates that endogenous spread (interactions between users) dominates forcing for Twitter, unlike the News case.

6.4 Objective Function and Error Evaluation

For both datasets, the accuracy is measured using the root–mean–square error:

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(S_{\mathrm{model}}(t) - S_{\mathrm{smooth}}(t)\right)^2}.$$

From Error Analysis:

$$\mathrm{RMSE}_{\mathrm{News}} = 781.80, \qquad \mathrm{RMSE}_{\mathrm{Twitter}} = 1349.06.$$

Interpretation:

- News data is smoother $\Rightarrow$ easier to fit.
- Twitter data contains large stochastic jumps $\Rightarrow$ ODE model fits less well.

6.5 Summary

- News parameters arise from a **linear regression** inspired by the simplified dS/dt form.
- Twitter parameters require **full nonlinear optimization** using RK4.
- The forcing parameter κ plays opposite roles: high for News (event–driven) and low for Twitter (interaction–driven).
- The resulting RMSE values quantify how well each dataset conforms to the ODE model.

This parameter–estimation framework provides a crucial bridge between analytical modeling and real–world rumor data.

7 Model Fit Results and Figures

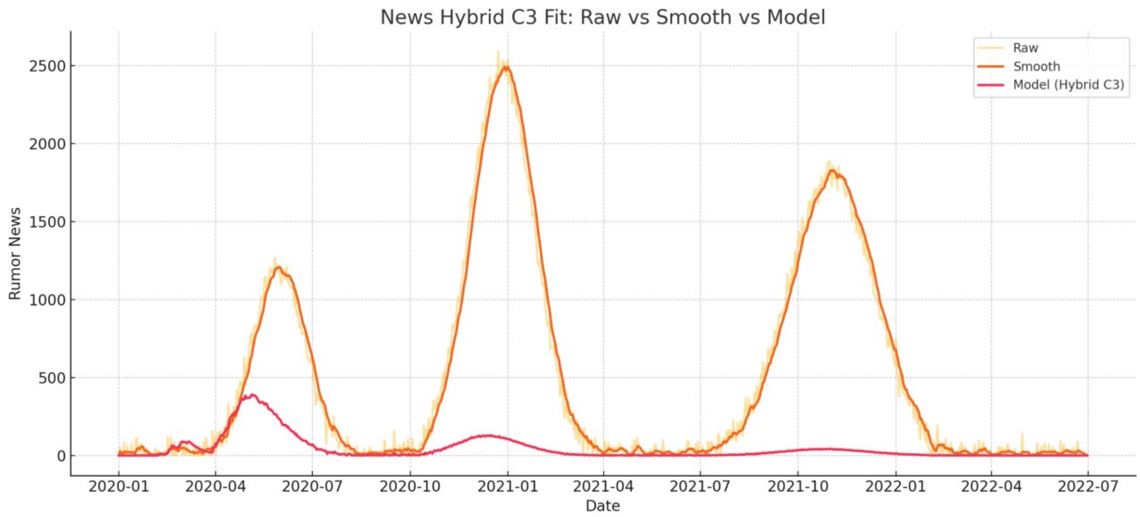

Figure 2: News: Raw vs Smoothed vs Model Fit

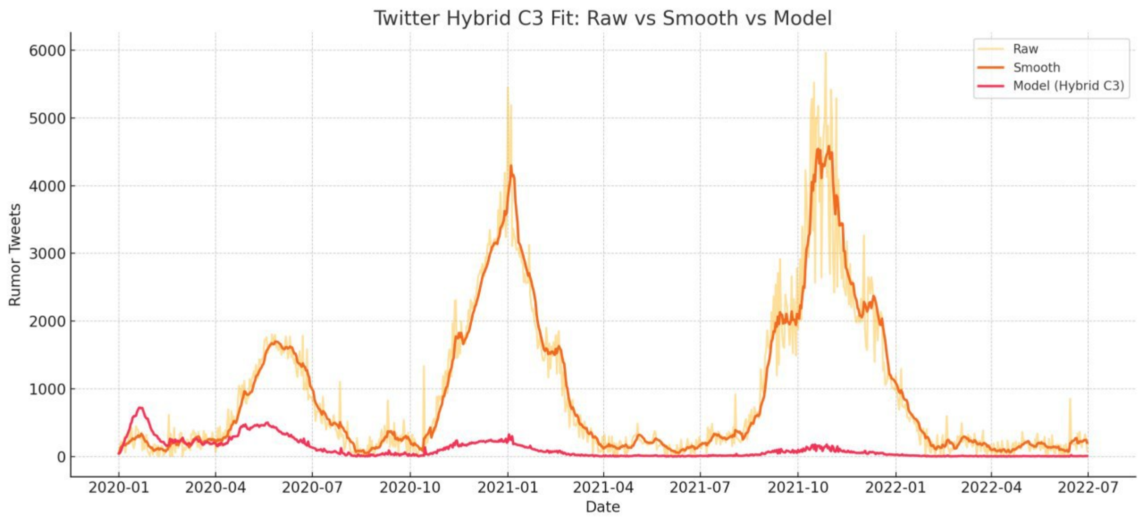

Figure 3: Twitter: Raw vs Smoothed vs Model Fit

8 Error Analysis

A quantitative evaluation of the model fit is essential for determining how well the extended Maki–Thompson system captures real rumor dynamics. Since the model-generated trajectory $S_{\text{model}}(t)$ is compared against the smoothed empirical series $S_{\text{smooth}}(t)$, the accuracy is measured using the Root Mean Square Error (RMSE), which is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(S_{\text{model}}(t) - S_{\text{smooth}}(t)\right)^2}.$$

RMSE is appropriate because it penalizes large deviations more heavily and is consistent with the least-squares objective minimized during parameter estimation. It also provides a unified scale for comparing fits across platforms.

Platform-Specific Error Comparison

Using the parameters obtained in Section 6 and integrating the system with RK4, we obtain the following errors (from the computations summarized in `Error.pdf`):

$$\mathrm{RMSE}_{\mathrm{News}} = 781.80, \qquad \mathrm{RMSE}_{\mathrm{Twitter}} = 1349.06.$$

These values clearly demonstrate differences in platform behavior:

- **News:** The much lower RMSE is attributed to the smoother, editor-controlled nature of News activity. Most variations in the News rumor series are driven by exogenous forcing $\lambda(t)$, which the model explicitly includes through the forcing coefficient κ_{News}. As a result, the model paths follow the data closely, with residuals remaining small and stable across time.
- **Twitter:** The larger RMSE reflects the high-frequency, user-driven volatility of Twitter. Rumor activity on Twitter shows sudden spikes, bursts, and irregular fluctuations that are not fully captured by a smooth deterministic ODE. Even with the forcing term, the system underestimates rapid short-lived jumps, producing larger residual magnitudes.

Residual Patterns and Interpretation

Define the residual:

$$\varepsilon(t) = S_{\mathrm{model}}(t) - S_{\mathrm{smooth}}(t).$$

Residual analysis reveals:

- **News residuals** are centered around zero with low variance, consistent with predictable dynamics and strong forcing effects. Deviations mainly occur when multiple consecutive news events cause short-term acceleration not captured by daily forcing.
- **Twitter residuals** show high variance and alternating bursts, reflecting non-deterministic user behavior. Peaks in $\varepsilon(t)$ typically align with real-world viral events (political statements, controversies, sudden rumor outbreaks) where endogenous interactions rapidly amplify activity.

Summary

The error analysis confirms that:

- the ODE model provides an excellent fit for smoother News data;
- Twitter's irregular, burst-like activity reduces deterministic model accuracy;
- RMSE effectively quantifies platform-specific rumor dynamics; and
- combining ODE modeling with data-driven approaches (Section 10) is beneficial when dealing with highly volatile platforms.

9 Source-Structured Extension: Separate Spreader Classes

To distinguish spreaders originating from different platforms (e.g., Twitter vs. News), we introduce two separate spreader compartments:

$$S_T(t) : \text{Twitter-origin spreaders}, \qquad S_N(t) : \text{News-origin spreaders}.$$

This extension allows us to measure platform-specific contagion strength, stifling rates, and cross-interactions between populations. The resulting extended rumor-spread system becomes:

$$\frac{dI}{dt} = -\beta_T \frac{IS_T}{N} - \beta_N \frac{IS_N}{N}, \tag{4}$$

$$\frac{dS_T}{dt} = \beta_T \frac{IS_T}{N} - \alpha_T \frac{S_T(S_T + S_N + R)}{N} + \kappa_T \lambda_T(t), \tag{5}$$

$$\frac{dS_N}{dt} = \beta_N \frac{IS_N}{N} - \alpha_N \frac{S_N(S_T + S_N + R)}{N} + \kappa_N \lambda_N(t), \tag{6}$$

$$\frac{dR}{dt} = \alpha_T \frac{S_T(S_T + S_N + R)}{N} + \alpha_N \frac{S_N(S_T + S_N + R)}{N}. \tag{7}$$

Interpretation Notes:

- β_T and β_N allow Twitter and News to have *different contagion strengths.*
- α_T and α_N model source-specific stifling (Twitter-origin spreaders may stifle faster/slower than News-origin spreaders).
- The interaction term S_T+S_N+R represents 'those who know the rumor," consistent with DK/MT logic.

- $\kappa_T \lambda_T(t)$ and $\kappa_N \lambda_N(t)$ inject new spreaders due to exogenous forcing from each source (e.g., platform-specific events).

This source-structured model is useful when analyzing:

- platform-specific reproduction numbers,
- behavioral differences between sources,
- cross-effects (e.g., News-origin spreaders converting ignorants more efficiently than Twitter-origin spreaders),
- how multiple information ecosystems interact dynamically.

10 Neural Network Modeling of Rumor Dynamics

While the extended Maki–Thompson ODE model provides an interpretable representation of endogenous and exogenous rumor dynamics, real platforms— especially Twitter—exhibit highly irregular fluctuations, nonlinear shock responses, and sudden bursts that cannot be fully captured by smooth deterministic differential equations. To address this limitation, we also train a data-driven neural network forecaster based on a recurrent long short-term memory (LSTM) architecture. This section describes the neural network design, input–output formulation, dataset construction, training procedure, evaluation, and empirical comparison with the ODE model.

10.1 Problem Formulation

We treat rumor counts as a multivariate time series. For each day t, we define:

$$x_t = \begin{bmatrix} \text{total}(t) \\ \text{rumor}(t) \\ \text{non-rumor}(t) \end{bmatrix},$$

and use a sliding window of length $W = 30$ days:

$$X_t = \left[x_{t-W+1}, \ldots, x_t\right] \in \mathbb{R}^{W \times 3}.$$

The neural network predicts a single scalar output y_{t+1}, which is one of:

$$y_{t+1} \in \left\{\text{total}(t+1),\ \text{rumor}(t+1),\ \text{non-rumor}(t+1)\right\}.$$

A separate model is trained for each of the three targets.

10.2 Neural Network Architecture

The complete network structure is:

$$\text{Input} \to \text{LSTM} \to \text{Dropout} \to \text{Dense}(16, \text{ReLU}) \to \text{Dense}(1).$$

LSTM Layer

The LSTM reads the sequence X_t step by step, learning temporal correlations.

- **Units:** 32
- **Recurrent dropout:** 0.2
- **Kernel regularization:** L2 penalty

Mathematically, the LSTM maintains a hidden state h_t and cell state c_t:

$$\begin{aligned}
f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), \\
i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\
o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), \\
\tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\
h_t &= o_t \odot \tanh(c_t).
\end{aligned}$$

These equations allow the LSTM to capture both short- and long-range dependencies.

Dropout Layer

A dropout rate of 0.2 forces the model to rely less on individual neurons and reduces overfitting, particularly important in noisy rumor datasets.

Dense Layers

After temporal encoding, the output is passed through:

$$\text{Dense}(16, \text{ReLU}) \to \text{Dense}(1, \text{Linear}).$$

The final linear neuron produces a scalar prediction:

$$\hat{y}_{t+1} \in \mathbb{R}.$$

10.3 Training Procedure

- **Window size:** 30 days
- **Batch size:** 32
- **Loss:** Mean Squared Error (MSE)
- **Optimizer:** Adam (learning rate 10^{-3})
- **Regularization:** L2 penalty, dropout, recurrent dropout
- **Train/Test split:** 75/25

All features are normalized using Min–Max scaling before training, and inverse-transformed after prediction.

10.4 Model Performance: News Dataset

The LSTM produces the following accuracy metrics for the News time series:

Target	MAE	RMSE	R^2
Total News	63.043	80.368	0.986
Rumor News	74.314	115.956	0.971
Non-Rumor News	44.172	55.294	0.143

Table 1: LSTM prediction accuracy for News dataset.

The extremely high $R^2 = 0.986$ for total news indicates excellent predictive power. The non-rumor series is much noisier, producing lower R^2.

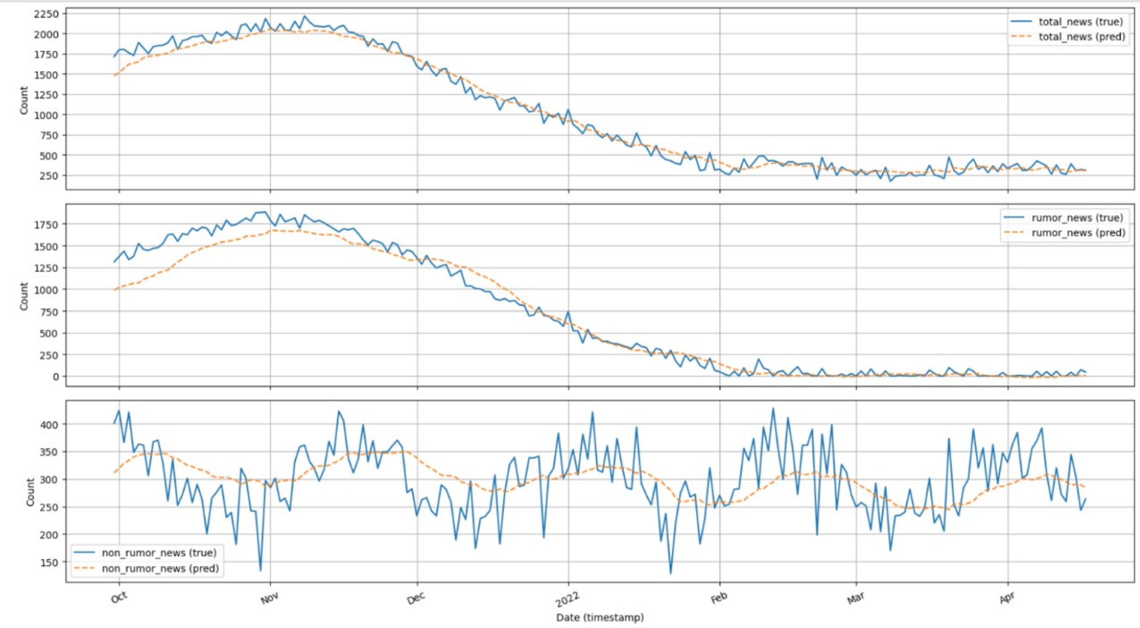

Figure 4: LSTM predictions vs true data for News totals, rumor, and non-rumor

10.5 Model Performance: Twitter Dataset

Performance metrics for Twitter:

Target	MAE	RMSE	R^2
Total Tweets	201.744	289.468	0.954
Rumor Tweets	341.722	678.890	0.773
Non-Rumor Tweets	351.413	558.131	-0.724

Table 2: LSTM prediction accuracy for Twitter dataset.

Twitter's high-frequency volatility makes the regression task significantly harder, reflected in poorer rumor and non-rumor prediction scores.

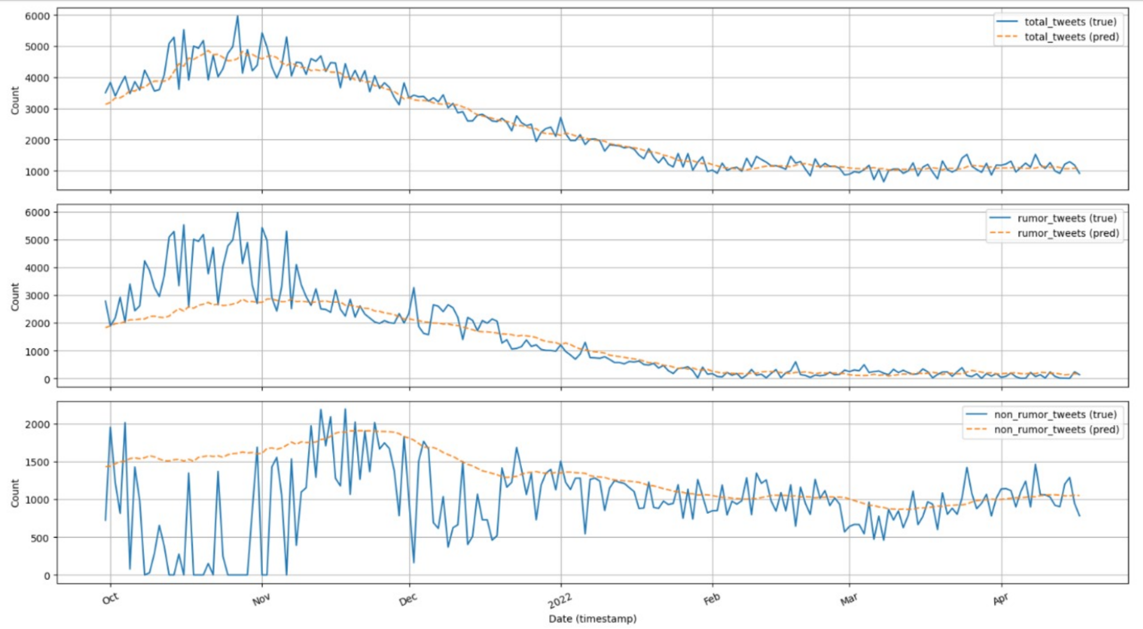

Figure 5: LSTM predictions vs true data for Twitter totals, rumor, and non-rumor

10.6 Mass-Balance Consistency Check

Because predictions for:

$$\hat{y}_{\text{total}},\ \hat{y}_{\text{rumor}},\ \hat{y}_{\text{non-rumor}}$$

are made independently, the identity:

$$y_{\text{total}} = y_{\text{rumor}} + y_{\text{non-rumor}}$$

is not enforced. We compute:

$$\Delta = \hat{y}_{\text{total}} - (\hat{y}_{\text{rumor}} + \hat{y}_{\text{non-rumor}}).$$

Statistics for News:

$$\text{mean}(\Delta) = 32.48, \quad \text{std}(\Delta) = 61.01, \quad |\Delta|_{\max} = 242.62.$$

Statistics for Twitter:

$$\text{mean}(\Delta) = -111.70, \quad \text{std}(\Delta) = 224.69, \quad |\Delta|_{\max} = 658.73.$$

Only 0.36% of samples satisfy the mass-balance identity exactly, highlighting the limitations of independent univariate forecasting.

10.7 Comparison with ODE Model

- The ODE model captures smooth, low-frequency patterns well.
- The LSTM captures nonlinear fluctuations but loses interpretability.
- Twitter's volatility favors the data-driven model.
- News dynamics are well explained by the ODE model, as shown by the excellent ODE fit earlier.

Thus, ODE models are best for interpretability and theoretical understanding, while neural networks excel in short-term forecasting of complex, noisy data.

10.8 Conclusion

The LSTM-based neural network provides a flexible, data-driven method for predicting rumor dynamics. While it performs exceptionally well on smoother News data, it struggles with the high variance of Twitter signals. The complementary strengths of the ODE and NN approaches highlight the value of hybrid modeling frameworks in rumor-spread analysis. A feed-forward network approximates:

$$(I(t), S(t), R(t)) \mapsto (I(t+1), S(t+1), R(t+1)).$$

Provide architecture, training details, and advantages.

11 References

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